

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457248.9132 +/- 0.007 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.041
Transit depth (Rp/Rs)^2	1.93 +/- 1.13 [%]
Orbital Inclination (inc)	89.68 +/- 1.43
Airmass coefficient 1 (a1)	0.049 +/- 0.032
Airmass coefficient 2 (a2)	2.77 +/- 0.99
Scatter in the residuals of the lightcurve fit is	64.17 %
Best Comparison Star	#3 - [174, 36]
Optimal Aperture	1.46
Optimal Annulus	3.0
Transit Duration (day)	0.1567 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.139 $\pm$ 0.041 vs 0.1036 (deviation 0.67 σ)
Duration fit vs published	0.1567 d vs 0.1567 d (deviation 0.0 σ)
Mid-time O-C	-15.7 min
Depth SNR	2.6
Residual scatter	64.17 %

Light curve

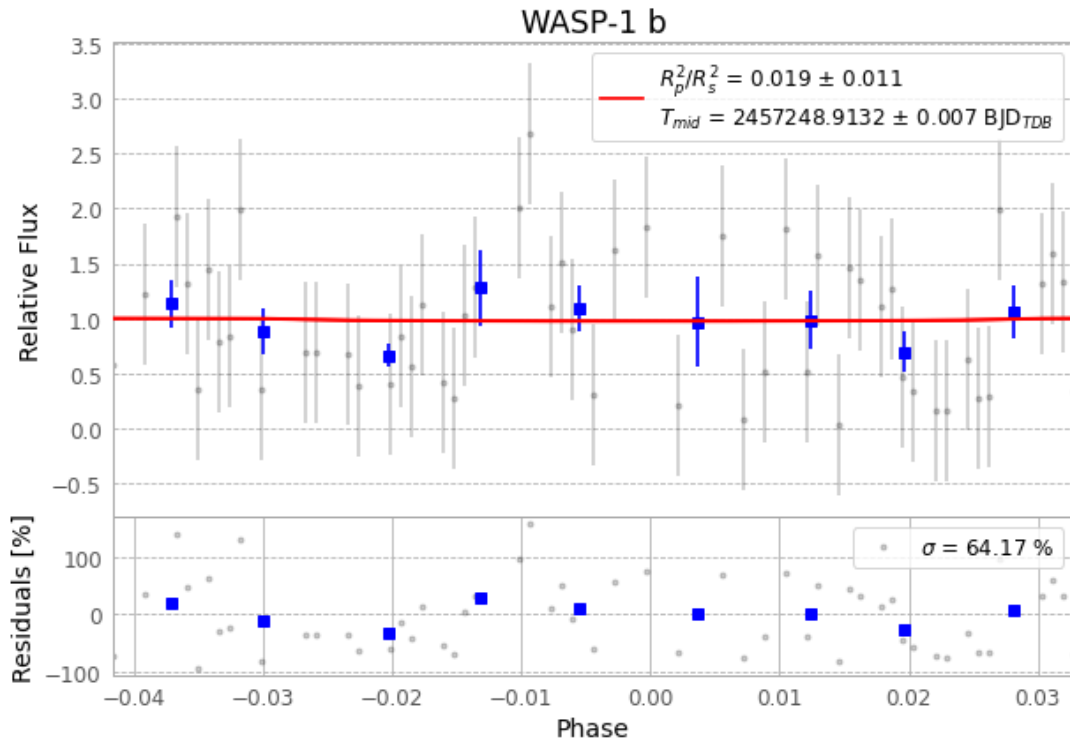


Figure 1 — FinalLightCurve_WASP-1 b_2015-08-14.png (SHA-256 e2cae73dbe44490e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [450, 427], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c673fb9244b3214d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

92 FITS frames (60 MB), WASP-1150814071812.FITS → WASP-1150814115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-150814.log (acff8d24f03a...), FinalLightCurve_WASP-1 b_2015-08-14.png (e2cae73dbe44...), FinalLightCurve_WASP-1 b_2015-08-14.csv (c276d1c485c1...)

Compute resources

Wall time 260.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 23:03Z.